Independent validation report: Retail Application Scorecard (RCS-APP-02)

Model under test: redeveloped/credit_scorecard/scorecard.py (Python rebuild, sha256 dc0f6c4e40ed, approved at Gate 3)
Generated: 2026-11-08T21:43:36.727305+00:00
Status: pending Gate 4

OVERALL RATING: AMBER - Fit for use with conditions: remediate the amber findings.

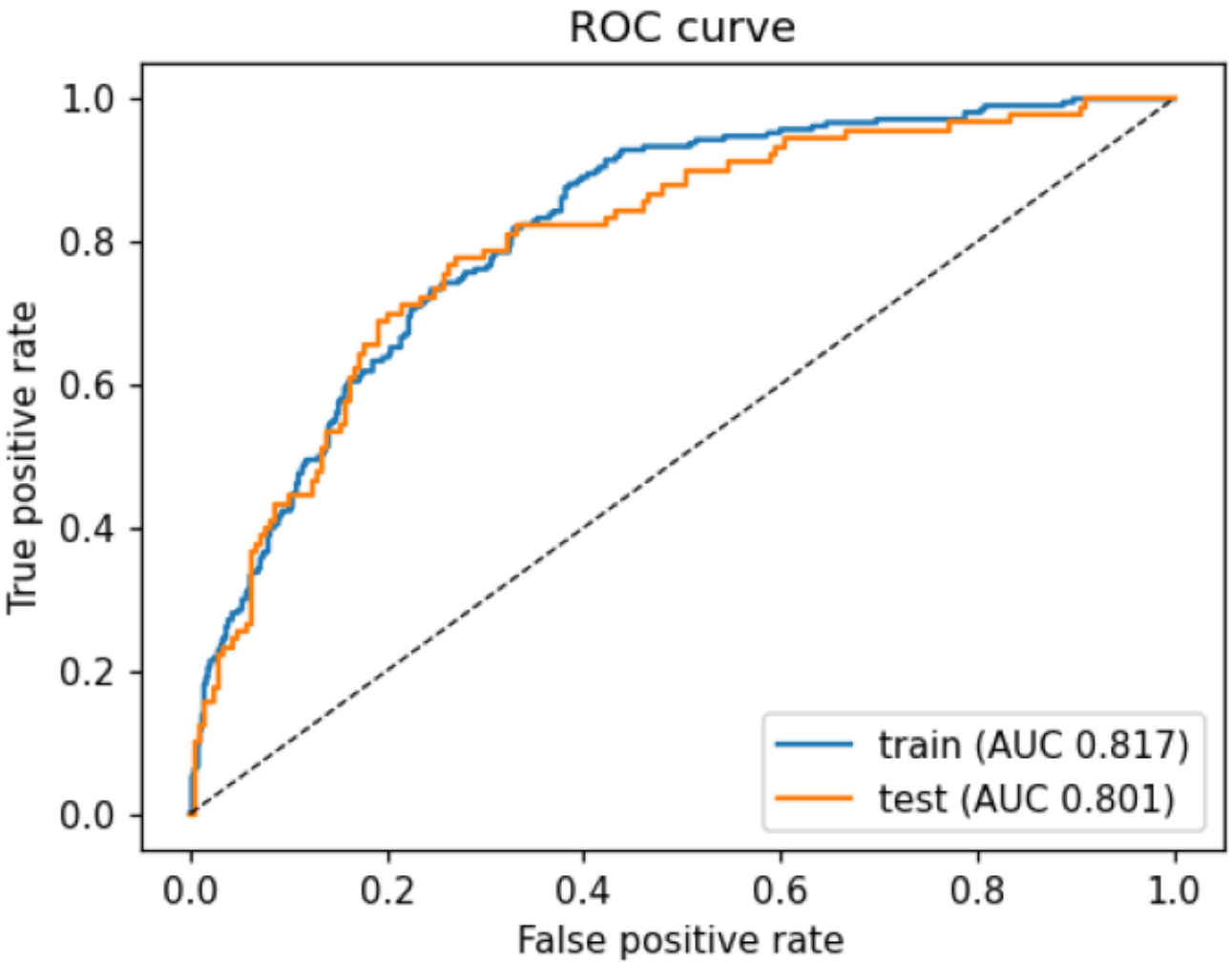
Ratings by area: performance GREEN, calibration GREEN, stability AMBER, fairness AMBER

Executive summary:
RCS-APP-02, a high-tier retail application PD scorecard, receives an overall amber rating. Discrimination and calibration are both green: test AUC is 0.801 with limited overfitting (AUC drop 0.016), and mean predicted default of 0.296 aligns with the observed rate of 0.300. The amber rating is driven by two areas: a synthetic out-of-time tranche showing score drift above the green threshold, and fairness evidence where the age-group equalized odds difference is amber and the foreign worker group is too small to test. The rebuild reproduces the Gate 3 model's test predictions exactly, so the conclusions apply to the approved implementation. All results rest on a single 300-row test sample and a synthetic time ordering, which limits the strength of the assurance.

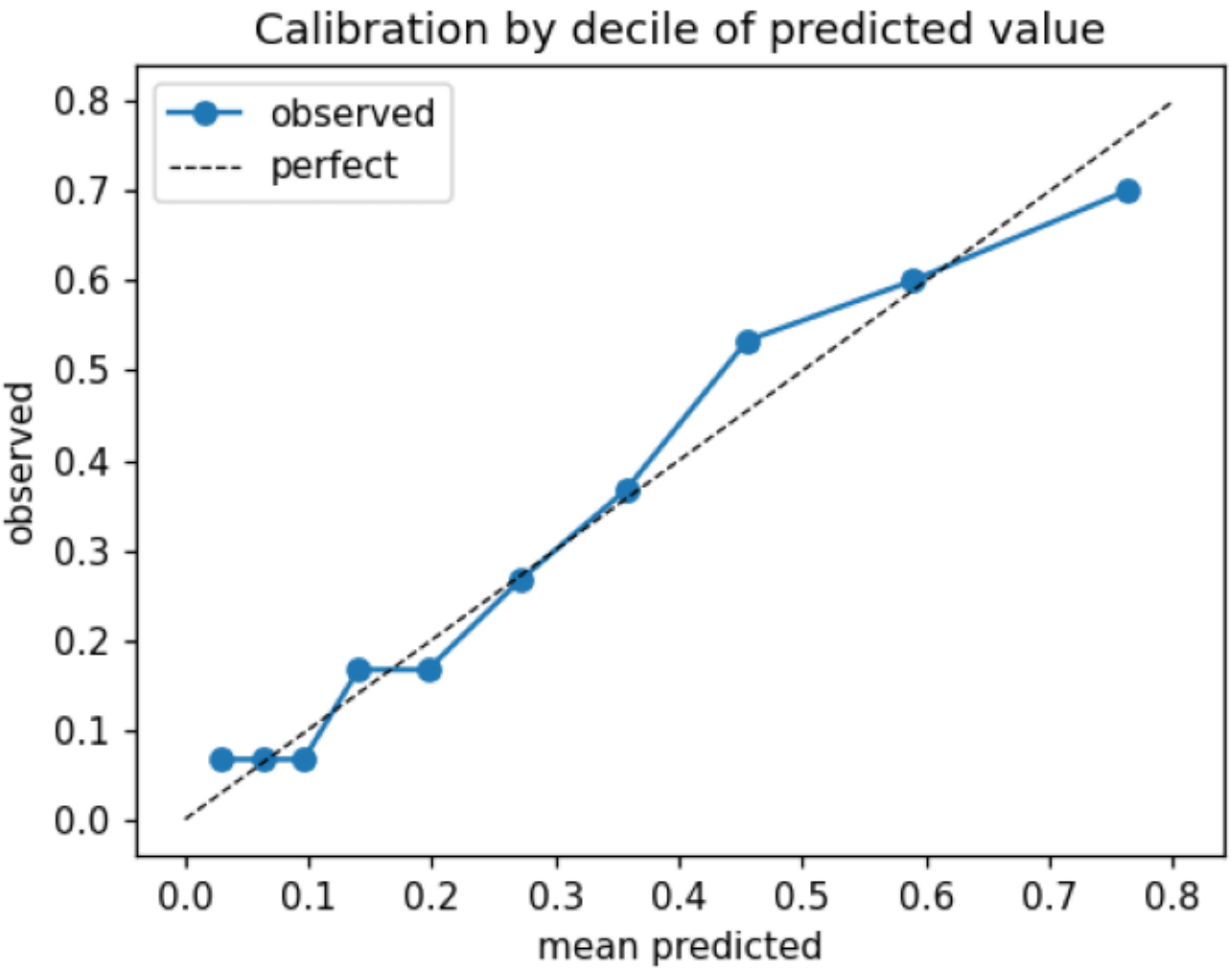
Checks:
[GREEN] performance: Discrimination: test AUC = 0.801 (>= 0.75 green, >= 0.65 amber) 95% bootstrap CI 0.745-0.853
[GREEN] performance: Overfitting: train AUC - test AUC = 0.016 (<= 0.03 green, <= 0.07 amber)
[GREEN] calibration: Calibration in the large: |mean predicted / observed - 1| = 0.013 (<= 0.1 green, <= 0.2 amber)
[GREEN] calibration: Hosmer-Lemeshow goodness of fit (p-value) = 0.900 (>= 0.05 green, >= 0.01 amber) 10 equal-count groups, 8 d.f.
[GREEN] stability: Score PSI (train -> test) = 0.011 (<= 0.1 green, <= 0.25 amber)
[GREEN] stability: Characteristic stability: largest CSI = 0.043 (<= 0.1 green, <= 0.25 amber) variable age
[AMBER] stability: Synthetic out-of-time: worst tranche score PSI = 0.130 (<= 0.1 green, <= 0.25 amber) 3 tranches of the test set in id order (synthetic time)
[GREEN] fairness: sex: adverse impact ratio of approval rates = 0.981 (>= 0.8 green, >= 0.7 amber)
[GREEN] fairness: sex: equalized odds difference = 0.068 (<= 0.1 green, <= 0.2 amber)
[GREEN] fairness: age_group: adverse impact ratio of approval rates = 0.923 (>= 0.8 green, >= 0.7 amber)
[AMBER] fairness: age_group: equalized odds difference = 0.132 (<= 0.1 green, <= 0.2 amber)
[AMBER] fairness: foreign_worker: group too small to rate = n/a (n >= 30 per group) A202 n=9

Findings:
[medium] The worst synthetic out-of-time tranche shows a score PSI of 0.130, above the green threshold, while train-to-test PSI is green. The tranches are an id-ordered split rather than true time periods, so the drift signal cannot be attributed to genuine temporal change. -> Run a true out-of-time back-test on dated application data before or shortly after deployment, and set up periodic score PSI and CSI monitoring with escalation triggers in the ongoing monitoring plan.
[medium] The age-group equalized odds difference is 0.132, above the green limit, and age is a direct model input. The foreign worker comparison could not be rated because one group has only 9 records. -> Model Development should investigate which age bands drive the gap, assess whether the age bins are justified by risk evidence, and test alternatives such as coarser banding or constraints. Fairness testing for foreign worker status should be repeated on a larger sample or with pooled data, and the age result should be tracked in ongoing monitoring.

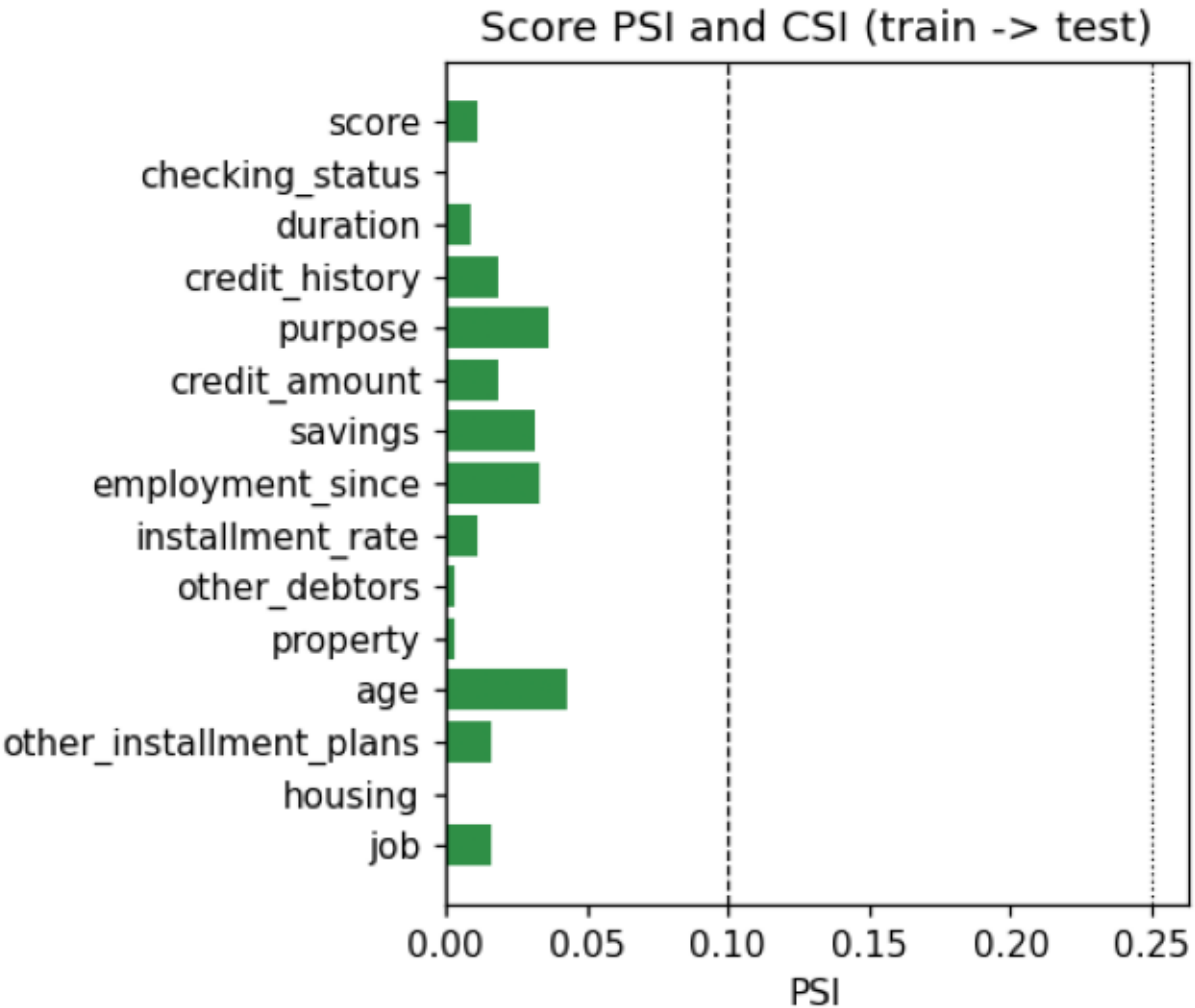
Performance: roc curve



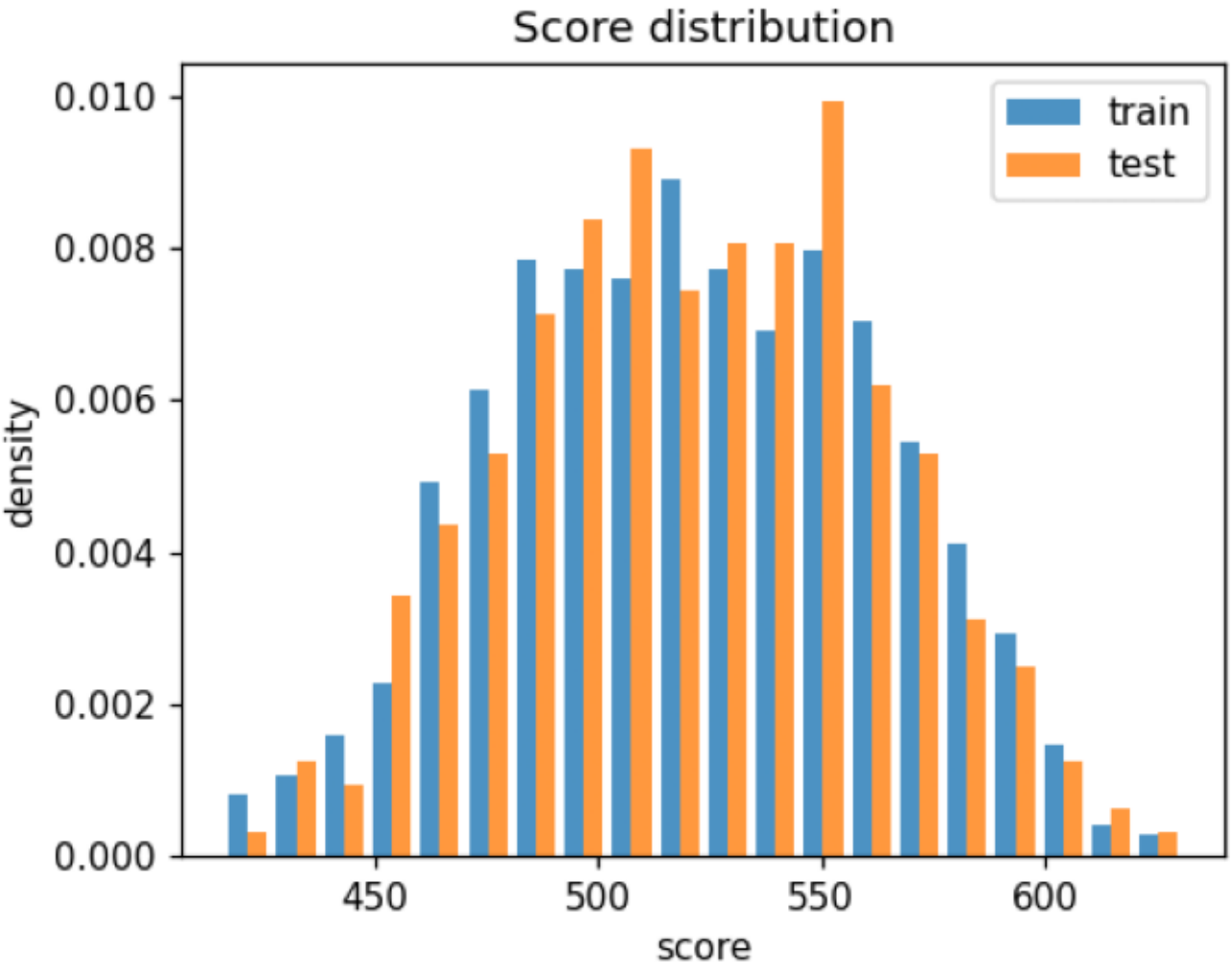
Calibration: calibration



Stability: psi by variable



Stability: score distribution



Fairness: approval rates

